

Aydın Buluç

Senior Libraries Engineer, NVIDIA

Berkeley, California

Former research page | [Google Scholar](#) | [LinkedIn](#)

Research and impact

High-performance computing, sparse linear algebra, and combinatorial scientific computing. Develops mathematically sound algorithms and scalable software for sparse matrices and tensors, distributed data structures, graph analysis, computational biology, and machine learning. Current work at NVIDIA focuses on GPU-accelerated sparse linear solvers, **primarily cuISS, as well as parts of cuDSS and cuSPARSE**.

Google Scholar: 10,350 citations; h-index 49; i10-index 112.

Metrics and publication ranking checked September 12, 2026.

Professional appointments

NVIDIA Corporation

Nov. 2025–present

Senior Libraries Engineer, Math Libraries. Sparse linear algebra, sparse linear solvers, and distributed sparse matrices.

Lawrence Berkeley National Laboratory

Apr. 2010–Nov. 2025

Senior Scientist (Aug. 2022–Nov. 2025); *Staff Scientist* (Mar. 2016–July 2022); *Research Scientist* (May 2012–Mar. 2016); *Luis W. Alvarez Postdoctoral Fellow* (Apr. 2010–Apr. 2012).

- Led PASSION Lab, a research team of UC Berkeley Ph.D. students and LBNL postdoctoral fellows, research scientists, and engineers.
- Directed **Sparsitute**, a DOE mathematical center spanning **six institutions and 30+ members**; set the technical agenda and coordinated research in sparse computation and software.
- LBNL PI for ExaGraph; contributed to ExaBiome protein-family identification and genome assembly at supercomputing scale; co-led the GraphBLAS effort with academic, laboratory, and industry collaborators.

University of California, Berkeley, EECS

Jan. 2017–Nov. 2025

Adjunct Associate Professor (July 2022–Nov. 2025); *Adjunct Assistant Professor* (Jan. 2017–June 2022). Modernized **CS267: Applications of Parallel Computers** with multiple lectures on parallel algorithms for machine learning and expanded coverage of distributed-memory programming, graph algorithms, sorting and searching, and computational biology.

Simons Institute for the Theory of Computing

Aug.–Dec. 2013

Visiting Scientist, Theoretical Foundations of Big Data Analysis program.

Education

Ph.D., Computer Science, UC Santa Barbara, 2010; advisor: John R. Gilbert. Thesis: *Linear Algebraic Primitives for Parallel Computing on Large Graphs*.

M.S., Computer Science, UC Santa Barbara, 2009.

B.S., Computer Science, Sabancı University, 2005; mathematics minor.

MIT Department of Mathematics, visiting student, 2008.

Honors and recognition

- **ACM Distinguished Member, 2025**, for contributions to high-performance algorithms and software.
- **ACM Gordon Bell Prize finalist, 2022**, team leader, “Extreme-scale many-against-many protein similarity search.”
- **IEEE TCSC Award for Excellence for Early Career Research, 2015.**
- **DOE Office of Science Early Career Award, 2013.**
- Euro-Par Distinguished Paper, 2018; SC Best Paper nominee, 2018 and 2021.
- HPCwire Best Use of HPC in Life Sciences, HipMer/MetaHipMer teams: Readers’ Choice, 2015; Editors’ Choice, 2021.

Selected research leadership

Sparsitute founding leadership. Lead PI and Center Director while at LBNL; DOE ASCR MMICC award, 2022–2027, \$13.5M across participating institutions (\$4.5M LBNL). Award dates describe the project period.

Selected PI awards. ExaGraph, institutional PI (FY2017–2023; \$2.74M LBNL); Scalable Graph Learning for Scientific Discovery (2021–2025; \$1.5M); Parallel Primitives for Randomized Algorithms on Sparse Data (2018–2021; \$1.5M); Energy-efficient Parallel Graph and Data Mining Algorithms (2013–2018; \$2.5M).

Mentoring

Ph.D. co-advising. Alok Tripathy (UC Berkeley, 2025; NVIDIA); Vivek Bharadwaj (UC Berkeley, 2025; Apple); Giulia Guidi (UC Berkeley, 2022; Cornell); Benjamin Brock (UC Berkeley, 2022; Intel Labs); Carl Yang (UC Davis, 2019; Waymo). Other recent advisees include Tianyu Liang, Yen-Hsiang Chang, and Gabriel Raulet.

Postdoctoral supervision and co-supervision. Helen Xu (Georgia Tech), Can Kizilkale (UC Berkeley), Prashant Pandey (Northeastern University), Israt Nisa (Amazon), Oguz Selvitopi (LBNL), Saliya Ekanayake (d-Matrix), Ariful Azad (Texas A&M), Sang-Yun Oh (UC Santa Barbara), and Harsha Vardhan Simhadri (Microsoft Research).

SIAM and professional service

Editorial and society service. Associate Editor, *SIAM Journal on Scientific Computing*; Founding Associate Editor, *ACM Transactions on Parallel Computing* (2013–2020). SIAM Membership Committee (2022–2024); George Pólya Prize for Mathematical Exposition Selection Committee (2025); SIAG/Linear Algebra Early Career Prize Committee (2027 award).

Conference leadership and education. SPAA Area Chair (2025); SC Algorithms Track Chair (2018), Applications Track Vice Chair (2017); SIAM CSC Publications Chair (2018). GraphBLAS tutorials at SIAM CSE and ICS (2021); minisymposia at SIAM PP (2024), CSE (2021, 2023), and ICIAM (2019); GrAPL steering committee (from 2019).

Selected software and standards

GraphBLAS C API and standards leadership. Made substantial technical contributions to the GraphBLAS C API as a member of the original five-person C API team. Also contributed to the strategic direction of the broader GraphBLAS effort through steering committee membership, helping connect interface design, algorithms, and implementations across academia, national laboratories, and industry.

HipMCL and PASTIS: from algorithms to biological discovery. Conceived and led the design of HipMCL, a scalable parallel implementation of Markov clustering, and PASTIS, a distributed-memory protein sequence similarity tool based on sparse matrix operations. Together, these tools helped more than double the number of known protein families in the global metagenomics study published in *Nature* in 2023 (publication 10).

Additional software contributions. Combinatorial BLAS, the GraphBLAST GPU graph framework, sparse matrix kernels and communication-avoiding methods, and HipMer/MetaHipMer genome assembly.

Twenty most-cited publications

Ordered by Google Scholar, September 12, 2026; citation counts in brackets. All authors listed using initials, except the CAMI consortium entry.

1. **A. Buluç**, H. Meyerhenke, I. Safro, P. Sanders, and C. Schulz. “Recent advances in graph partitioning.” In *Algorithm Engineering: Selected Results and Surveys*, pp. 117–158, 2016. [752]
2. **A. Buluç**, J. T. Fineman, M. Frigo, J. R. Gilbert, and C. E. Leiserson. “Parallel sparse matrix-vector and matrix-transpose-vector multiplication using compressed sparse blocks.” *ACM SPAA*, pp. 233–244, 2009. [648]
3. **A. Buluç** and J. R. Gilbert. “The Combinatorial BLAS: Design, implementation, and applications.” *International Journal of High Performance Computing Applications*, 25(4):496–509, 2011. [554]
4. CAMI consortium (including **A. Buluç**). “Critical Assessment of Metagenome Interpretation: the second round of challenges.” *Nature Methods*, 19(4):429–440, 2022. [458]
5. J. Kepner, P. Aaltonen, D. Bader, **A. Buluç**, F. Franchetti, J. Gilbert, D. Hutchison, M. Kumar, A. Lumsdaine, H. Meyerhenke, S. McMillan, J. Moreira, J. Owens, C. Yang, M. Zalewski, and T. Mattson. “Mathematical foundations of the GraphBLAS.” *IEEE HPEC*, pp. 1–9, 2016. [379]
6. **A. Buluç** and K. Madduri. “Parallel breadth-first search on distributed memory systems.” *SC*, 2011. [360]
7. J. A. Chapman, M. Mascher, **A. Buluç**, K. Barry, E. Georganas, A. Session, V. Strnadova, J. Jenkins, S. Sehgal, L. Olikar, J. Schmutz, K. A. Yelick, U. Scholz, R. Waugh, J. A. Poland, G. J. Muehlbauer, N. Stein, and D. S. Rokhsar. “A whole-genome shotgun approach for assembling and anchoring the hexaploid bread wheat genome.” *Genome Biology*, 16:26, 2015. [327]

Twenty most-cited publications (continued)

8. **A. Buluç** and J. R. Gilbert. “Parallel sparse matrix-matrix multiplication and indexing: Implementation and experiments.” *SIAM Journal on Scientific Computing*, 34(4):C170–C191, 2012. [302]
9. **A. Buluç** and J. R. Gilbert. “On the representation and multiplication of hypersparse matrices.” *IPDPS*, 2008. [298]
10. G. A. Pavlopoulos, F. A. Baltoumas, S. Liu, O. Selvitopi, A. P. Camargo, S. Nayfach, A. Azad, S. Roux, L. Call, N. N. Ivanova, I.-M. Chen, D. Paez-Espino, E. Karatzas, Novel Metagenome Protein Families Consortium, I. Iliopoulos, K. Konstantinidis, J. M. Tiedje, J. Pett-Ridge, D. Baker, A. Visel, C. A. Ouzounis, S. Ovchinnikov, **A. Buluç**, and N. C. Kyrpides. “Unraveling the functional dark matter through global metagenomics.” *Nature*, 622:594–602, 2023. [248]
11. A. Azad, **A. Buluç**, and J. Gilbert. “Parallel triangle counting and enumeration using matrix algebra.” *IEEE IPDPS Workshops*, pp. 804–811, 2015. [226]
12. A. Azad, G. A. Pavlopoulos, C. A. Ouzounis, N. C. Kyrpides, and **A. Buluç**. “HipMCL: A high-performance parallel implementation of the Markov clustering algorithm for large-scale networks.” *Nucleic Acids Research*, 46(6):e33, 2018. [224]
13. C. Yang, **A. Buluç**, and J. D. Owens. “Design principles for sparse matrix multiplication on the GPU.” *Euro-Par*, pp. 672–687, 2018. [190]
14. C. Yang, **A. Buluç**, and J. D. Owens. “GraphBLAST: A high-performance linear algebra-based graph framework on the GPU.” *ACM Transactions on Mathematical Software*, 48(1):1–51, 2022. [188]
15. **A. Buluç** and J. R. Gilbert. “Challenges and advances in parallel sparse matrix-matrix multiplication.” *ICPP*, pp. 503–510, 2008. [188]
16. **A. Buluç**, S. Williams, L. Oliker, and J. Demmel. “Reduced-bandwidth multithreaded algorithms for sparse matrix-vector multiplication.” *IPDPS*, pp. 721–733, 2011. [183]
17. A. Tripathy, K. Yelick, and **A. Buluç**. “Reducing communication in graph neural network training.” *SC*, 2020. [182]
18. **A. Buluç**, J. R. Gilbert, and C. Budak. “Solving path problems on the GPU.” *Parallel Computing*, 36(5–6):241–253, 2010. [174]
19. A. Azad, G. Ballard, **A. Buluç**, J. Demmel, L. Grigori, O. Schwartz, S. Toledo, and S. Williams. “Exploiting multiple levels of parallelism in sparse matrix-matrix multiplication.” *SIAM Journal on Scientific Computing*, 38(6):C624–C651, 2016. [153]
20. T. Mattson, D. Bader, J. Berry, **A. Buluç**, J. Dongarra, C. Faloutsos, J. Feo, J. Gilbert, J. Gonzalez, B. Hendrickson, J. Kepner, C. Leiserson, A. Lumsdaine, D. Padua, S. Poole, S. Reinhardt, M. Stonebraker, S. Wallach, and A. Yoo. “Standards for graph algorithm primitives.” *IEEE High Performance Extreme Computing Conference (HPEC)*, pp. 1–2, 2013. [148]